

Supplement: LLM Chart Judges Reward Labels, Audiences Reward Topics

All tables are generated by `src/11_supplement.py` from the CSVs in `analysis/`. Pairs and prompts are identical to the main paper unless stated.

S1. Data cleaning

Filters in order (`src/02_clean.py`). Deviations from our original plan: (i) a new filter for blank images (grayscale SD < 2), which are video thumbnails labeled from their titles; (ii) the perceptual-hash duplicate threshold was tightened from Hamming 4 to 2, because visual QA showed that distance-4 matches were different charts with shared layouts; (iii) a new rule drops groups of 3 or more near-duplicates whose titles are unrelated (median token-set similarity < 60), which removes link previews and site banners; (iv) transparent images are composited on white. No rule uses votes.

Table 1: Cleaning funnel.

Step	Rows remaining	Removed
raw rows	52836	0
unreadable image, missing score or date	52836	0
blank image (grayscale SD < 2)	52688	148
chart type empty / none / Other	46239	6449
short side < 300 px	44433	1806
score < 5	35928	8505
near-duplicate (pHash ≤ 2) or placeholder group	34556	1372

S2. Other prompts, title-only control, close pairs

P2 is a six-dimension VisJudge-style rubric (1–5 each, averaged). P3 is a pairwise prompt run in both orders: credit 1 if both orders pick the winner, 0 if both pick the loser, 0.5 if they disagree. The title-only control asks gpt-5 which of two titles got more upvotes, in both orders. These rows come from the earlier, smaller runs: P3 on the first 250 or 500 clear pairs, P2 on the first 150, and close pairs (vote ratio 1.5–2.5) on 200. Δ is the judge minus the image-blind baseline on the same pairs (95% bootstrap CI).

S3. Calibration on VisJudge-Bench

150 single-view items sampled with seed 20261008. P2 rubric mean vs. expert `overall_score`. Published values (VisJudge-Bench v3, 648-item test split over all types): GPT-5 $r = 0.428$, MAE 0.553. The test split cannot be reconstructed from the released file, so the comparison is approximate.

S4. Memorization

Training cutoffs: gpt-5 2024-09-30, Claude Haiku 4.5 Feb 2025 (reliable), Gemini 2.5 Flash Jan 2025. The dataset ends Jan 2025, so only gpt-5 has post-cutoff pairs. The recognition probe asked each judge whether it had seen each

Table 2: Rubric (P2), pairwise (P3), title-only, and close-pair results.

Set	Row	n	Acc.	Δ	lo	hi	Order flips	Picks first
clear	Gemini 2.5 Flash P3	500	0.564	-0.042	-0.089	+0.007	0.53	0.77
clear	gpt-5 title only	500	0.582	-0.024	-0.074	+0.024	0.50	0.26
clear	gpt-5 P3	250	0.636	+0.052	-0.028	+0.126	0.18	0.54
clear	Claude Haiku 4.5 P3	250	0.564	-0.020	-0.102	+0.064	0.18	0.42
clear	gpt-5 P2	150	0.600	+0.080	-0.023	+0.183	—	—
close	Gemini 2.5 Flash P1	200	0.512	+0.022	-0.068	+0.113	—	—
clear	feature baseline (OOF)	500	0.606	—	—	—	—	—
close	feature baseline (OOF)	200	0.490	—	—	—	—	—

Table 3: Prompt contrasts. The rubric and pairwise formats do not beat holistic P1, and images do not beat titles for gpt-5.

Paired contrast (same pairs)	n	Diff.	lo	hi
gpt-5 P2 minus gpt-5 P1	150	-0.043	-0.103	+0.013
gpt-5 P3 minus gpt-5 P1	250	+0.000	-0.048	+0.046
Claude Haiku 4.5 P3 minus Claude Haiku 4.5 P1	250	-0.020	-0.080	+0.032
Gemini 2.5 Flash P3 minus Gemini 2.5 Flash P1	500	-0.023	-0.054	+0.006
gpt-5 P3 minus gpt-5 title only	250	+0.008	-0.062	+0.074

of the 50 most-voted charts and, if so, for the title. Because 72% of these charts print their title, recalls that match the chart’s own OCR text are not counted as memory.

S5. Pre-registration (verbatim from the project log, written before the 2,000-pair run; names redacted)

- **Primary analysis:** prompt P1 (unchanged from src/prompts.py) on **all 2,000 pairs of pairs/clear.parquet**, for gpt-5, Claude Haiku 4.5 and Gemini 2.5 Flash, with the same settings as before. Pair credit is the same as before: 1 if the winner has the higher score, 0.5 on a tie or a failure, 0 otherwise.
- **Comparator:** the image-blind feature baseline’s out-of-fold predictions (analysis/04_feature_baseline_oof_clear.csv, computed 2026-10-08 and not refit).
- **Equivalence test:** TOST on the paired per-pair accuracy difference (judge minus baseline) with **margin ± 4 accuracy points**, $\alpha = 0.05$.
- **Robustness check:** the same analysis on pairs 501-2,000 only (the 1,500 pairs no judge had been run on before today).
- **Frontier judge:** Claude Sonnet 5.5 (global.anthropic.claude-sonnet-5-5), P1 only, on **the first 1,000 clear pairs** (2,000 images); the budget can’t cover all 2,000.
- **Secondary analyses** (same 2,000 pairs, P1): the added-value likelihood-ratio test and the text/OCR bias probe. Bias features need OCR on the 3,000 new images.
- **Nothing else changes:** prompts, pair sets, image preprocessing and judge settings for the three existing judges all stay as they were.
- **Budget mechanics:** HARD_STOP_USD is raised to 29.20. Spend is now re-read from disk every 50 new calls, so parallel processes see each other’s spend.
- **Addition, logged before any rerun result was seen:** P1 on the 150 VisJudge-Bench calibration items for all 4 judges (about \$0.80). This lets the expert-vs-audience contrast use the same prompt and the same metric (Spearman of the P1 score with expert overall_score vs with the audience month percentile). It addresses the review point that Spearman 0.09-0.12 against Pearson 0.41-0.47 compared different metrics and different prompts.

Table 4: Robustness check: P1 equivalence on the 1,500 pairs no judge saw during development.

Comparison (P1, 1,500 fresh pairs)	n	Diff.	90% lo	90% hi	TOST p	Verdict
gpt-5 P1 minus baseline	1500	-0.0277	-0.0553	+0.0000	0.233	inconclusive
Claude Haiku 4.5 P1 minus baseline	1500	-0.0317	-0.0587	-0.0047	0.305	inconclusive
Gemini 2.5 Flash P1 minus baseline	1500	-0.0140	-0.0403	+0.0127	0.055	inconclusive

Table 5: P2 calibration against expert ratings.

Judge	n	Pearson r	lo	hi	MAE
Gemini 2.5 Flash	150	0.472	0.319	0.599	0.871
gpt-5	150	0.414	0.262	0.537	0.622
Claude Haiku 4.5	150	0.467	0.326	0.590	0.692

- **Change to the addition above, logged before running it:** Sonnet 5.5 cost about 15% more per image than its smoke test showed. Total spend after the rerun is about \$28.35, and P1 calibration for all 4 judges (about \$1.00) would cross the \$29.20 hard stop. P1 calibration therefore runs for gpt-5, Haiku 4.5 and Gemini 2.5 Flash only (about \$0.57). Sonnet 5.5's expert agreement is not measured.

Table 6: Same prompt (P1), same metric (Spearman), different targets. Sonnet 5.5 was not run on calibration items because of the budget cap.

Judge	P1 ρ vs. experts	Audience images	P1 ρ vs. audience pct	lo	hi
gpt-5	0.442	4000	0.084	0.051	0.112
Claude Haiku 4.5	0.489	3999	0.115	0.086	0.145
Gemini 2.5 Flash	0.453	4000	0.133	0.103	0.164
Claude Sonnet 5.5	–	2000	0.119	0.077	0.164

Table 7: Accuracy by training-cutoff group (first 500 clear pairs).

Judge	Prompt	Cutoff group	n	Acc.	lo	hi
gpt-5	P1	post	37	0.581	0.446	0.703
gpt-5	P1	pre	463	0.579	0.537	0.619
gpt-5	P3	post	17	0.706	0.500	0.882
gpt-5	P3	pre	233	0.631	0.569	0.687
Claude Haiku 4.5	P1	pre	500	0.555	0.520	0.590
Claude Haiku 4.5	P3	pre	250	0.564	0.512	0.620
Gemini 2.5 Flash	P1	pre	500	0.587	0.555	0.618
Gemini 2.5 Flash	P3	pre	500	0.564	0.535	0.593

Table 8: Recognition probe.

Judge	n	Claims seen	Title recall (raw)	Recall not in image
Gemini 2.5 Flash	50	0.62	0.20	0.02
gpt-5	50	0.06	0.02	0.00
Claude Haiku 4.5	50	0.20	0.10	0.02